The First and Second Derivative Tests for Local Extrema

A Real-Analysis Treatment

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1 Introduction

Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. This essay treats two classical sufficient conditions for a differentiable function to have a local extremum at a point — the first and second derivative tests — with full statements of hypotheses and proofs, rather than as computational recipes.

Stated as a recipe, the tests reduce to: solve $f'(x) = 0$, inspect f' or f'', and declare a maximum or minimum. This conceals several logical subtleties that this essay makes explicit and proves:

1. $f'(c) = 0$ is necessary for a differentiable interior extremum, but not sufficient.
2. An extremum can occur where f' does not exist.
3. The condition $f''(c) > 0$ implies a local minimum only under the additional hypothesis $f'(c) = 0$, and even then $f''(c) = 0$ settles nothing.
4. Boundary points of the domain require separate treatment, since the tests are stated for interior points.

We also develop concavity precisely, since it is the concept underlying the informal language of a graph "bending upward" or "bending downward" used to motivate the second derivative test.

2 Local Extrema and Critical Points

Definition 2.1 (Local extremum)**.** Let $f : D \to \mathbb{R}$ and $c \in D$. We say f has a *local maximum* at c if

$$\exists\, \delta > 0 \quad \forall x \in D : \quad |x - c| < \delta \implies f(x) \leq f(c).$$

We say f has a *local minimum* at c if

$$\exists\, \delta > 0 \quad \forall x \in D : \quad |x - c| < \delta \implies f(x) \geq f(c).$$

A *local extremum* is a point at which f has a local maximum or a local minimum. The extremum is *strict* if the inequality is strict for every $x \neq c$ satisfying $|x - c| < \delta$.

Definition 2.2 (Critical point)**.** A point $c \in D$ is a *critical point* of f if either $f'(c) = 0$ or $f'(c)$ fails to exist.

Critical points are candidates for extrema; the definition asserts nothing about whether c actually is one.

Example 2.3. Let $f(x) = x^3$. Then $f'(0) = 0$, so 0 is a critical point, but for every $\delta > 0$ there exist points of $(-\delta, 0)$ with $f < 0 = f(0)$ and points of $(0, \delta)$ with $f > 0 = f(0)$. Hence 0 is neither a local maximum nor a local minimum.

Example 2.4. Let $f(x) = |x|$. Then $f'(0)$ does not exist, so 0 is a critical point, and $f(x) = |x| \geq 0 = f(0)$ for every x, so 0 is a (strict) local minimum.

These two examples show that neither implication

$$f'(c) = 0 \implies c \text{ is an extremum} \qquad \text{nor} \qquad c \text{ is an extremum} \implies f'(c) = 0$$

holds in general; the first fails by x^3, the second by $|x|$.

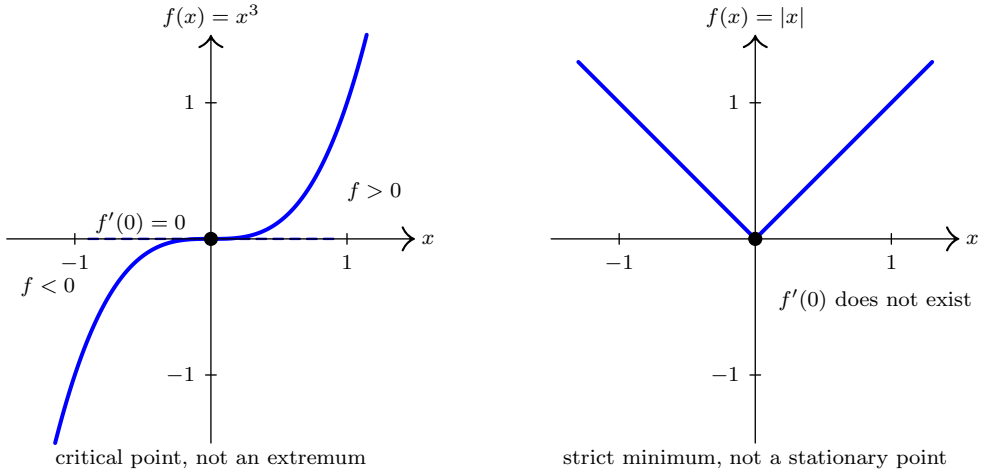


Figure 1: A critical point need not be an extremum (x^3), and an extremum need not be a stationary point ($|x|$).

3 Fermat's Theorem: A Necessary Condition

Theorem 3.1 (Fermat)**.** *Let $f : D \to \mathbb{R}$, let c be an interior point of D (i.e. $(c-\varepsilon, c+\varepsilon) \subseteq D$ for some $\varepsilon > 0$), and suppose f is differentiable at c and has a local extremum at c. Then $f'(c) = 0$.*

Proof. Suppose f has a local maximum at c; the minimum case is symmetric. Choose $\delta \in (0, \varepsilon)$ such that $f(x) \leq f(c)$ for all x with $|x - c| < \delta$. For $x \in (c, c + \delta)$, both $f(x) - f(c) \leq 0$ and $x - c > 0$, so

$$\frac{f(x) - f(c)}{x - c} \leq 0.$$

Since f is differentiable at c, the one-sided limit as $x \to c^+$ exists and equals $f'(c)$, so $f'(c) \leq 0$. For $x \in (c - \delta, c)$, both $f(x) - f(c) \leq 0$ and $x - c < 0$, so

$$\frac{f(x) - f(c)}{x - c} \geq 0,$$

and letting $x \to c^-$ gives $f'(c) \geq 0$. Together, $f'(c) \leq 0$ and $f'(c) \geq 0$ force $f'(c) = 0$. □

Both hypotheses are needed. Restricting $f(x) = x$ to $D = [0, 1]$, the point $c = 0$ is not interior to D; f has a local minimum there, yet the one-sided derivative at 0 is $1 \neq 0$. And $f(x) = |x|$ has a local minimum at the interior point 0, where f fails to be differentiable.

Fermat's theorem gives a necessary condition,

$$\text{differentiable interior extremum} \implies f'(c) = 0,$$

and the converse is false:

$$f'(c) = 0 \quad \not\Longrightarrow \quad c \text{ is a local extremum.}$$

The derivative tests below supply additional hypotheses under which a critical point *can* be classified.

4 The First Derivative Test

Theorem 4.1 (First derivative test)**.** *Let $f : D \to \mathbb{R}$, let c be an interior point of D, and suppose f is continuous at c and there exists $\delta > 0$ such that f is differentiable on $(c-\delta, c) \cup (c, c+\delta)$.*

1. *If $f' < 0$ on $(c-\delta, c)$ and $f' > 0$ on $(c, c+\delta)$, then f has a strict local minimum at c.*
2. *If $f' > 0$ on $(c-\delta, c)$ and $f' < 0$ on $(c, c+\delta)$, then f has a strict local maximum at c.*
3. *If f' has the same sign throughout $(c-\delta, c) \cup (c, c+\delta)$, then f has no local extremum at c.*

Proof. We prove (1); the proof of (2) is obtained by replacing f with $-f$.

Let $x \in (c-\delta, c)$. Since f is differentiable on $(c-\delta, c)$ and continuous at c, f is continuous on $[x, c]$ and differentiable on (x, c), so the Mean Value Theorem gives $\xi \in (x, c)$ with

$$\frac{f(c) - f(x)}{c - x} = f'(\xi).$$

Since $\xi \in (c-\delta, c)$, $f'(\xi) < 0$, and $c - x > 0$, so $f(c) - f(x) < 0$, i.e. $f(x) > f(c)$.

Let $x \in (c, c+\delta)$. Likewise the Mean Value Theorem on $[c, x]$ gives $\eta \in (c, x)$ with

$$\frac{f(x) - f(c)}{x - c} = f'(\eta) > 0,$$

so $f(x) > f(c)$.

Hence $f(x) > f(c)$ for every $x \in (c-\delta, c+\delta) \setminus \{c\}$, which is a strict local minimum.

For (3), suppose $f' > 0$ throughout $(c-\delta, c) \cup (c, c+\delta)$ (the case $f' < 0$ is symmetric). The same Mean Value Theorem argument on $[x, c]$ or $[c, x]$, for x on either side of c, shows f is strictly increasing on $(c-\delta, c+\delta)$; in particular $f(x) < f(c)$ for $x < c$ and $f(x) > f(c)$ for $x > c$ in this interval, so c is neither a local maximum nor a local minimum. □

The hypothesis doing the real work is not $f'(c) = 0$ but a *sign change* of f' across c; the theorem does not even require f to be differentiable at c itself, only continuous there.

5 Examples of the First Derivative Test

Example 5.1 (Local minimum)**.** For $f(x) = x^2$, $f'(x) = 2x$, which is negative on $(-\infty, 0)$ and positive on $(0, \infty)$. By the first derivative test, 0 is a strict local minimum.

Example 5.2 (Local maximum)**.** For $f(x) = -x^2$, $f'(x) = -2x$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$, so 0 is a strict local maximum.

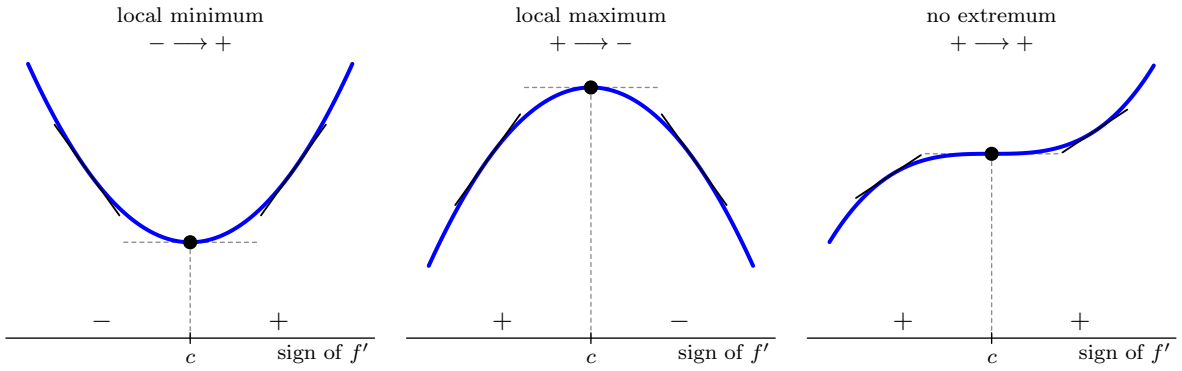


Figure 2: The first derivative test reads the sign of f' on each side of c; only a change of sign produces an extremum.

Example 5.3 (Critical point with no sign change)**.** For $f(x) = x^3$, $f'(x) = 3x^2 > 0$ for every $x \neq 0$; the sign does not change across 0, so 0 is not a local extremum, consistent with §2.

Example 5.4 (Extremum at a point of non-differentiability)**.** For $f(x) = |x|$, $f'(x) = -1$ on $(-\infty, 0)$ and $f'(x) = 1$ on $(0, \infty)$; f is continuous at 0 although $f'(0)$ does not exist. The hypotheses of the first derivative test are satisfied with $c = 0$, and the sign change gives a local minimum. This is the key advantage of the first derivative test over the second: it does not require differentiability at c.

6 Limits of the First Derivative Test

Remark 6.1 (No fixed-sign neighborhood). The theorem requires the existence of a single $\delta > 0$ on which f' has constant sign on each side of c. This need not hold. Let

$$f(x) = \begin{cases} x^4\big(2 + \sin(1/x)\big), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Since $2 + \sin(1/x) \geq 1$, $f(x) \geq x^4 > 0 = f(0)$ for every $x \neq 0$, so 0 is a strict local minimum. For $x \neq 0$,

$$f'(x) = 4x^3\big(2 + \sin(1/x)\big) - x^2\cos(1/x).$$

At $x = \pm 1/(2k\pi)$, where $\cos(1/x) = 1$, this equals $x^2(8x - 1)$, which is negative once $|x| < 1/8$; at $x = \pm 1/((2k+1)\pi)$, where $\cos(1/x) = -1$, it equals $x^2(8x + 1)$, which is positive once $|x| < 1/8$. Both families of points accumulate at 0 from each side, so f' changes sign infinitely often on every interval $(0, \delta)$ and $(-\delta, 0)$. No $\delta > 0$ makes f' of constant sign on either side, so the theorem's hypotheses are not met at $c = 0$ and the sign-chart method gives no direct conclusion, even though 0 is a strict local minimum.

Remark 6.2 (Continuity at c cannot be dropped). The hypothesis that f be continuous *at* c — not merely differentiable on the punctured neighborhood — is essential, since the theorem's conclusion is about the value $f(c)$ itself. Let

$$f(x) = \begin{cases} x^2, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Here $f'(x) = 2x$ is negative for $x < 0$ and positive for $x > 0$, the same pattern as in the local-minimum case of the theorem. But f is not continuous at 0: $f(0) = 1$ while $f(x) \to 0$ as $x \to 0$. Consequently $f(x) < f(0)$ for all $x \neq 0$ near 0, so 0 is actually a strict local *maximum*.

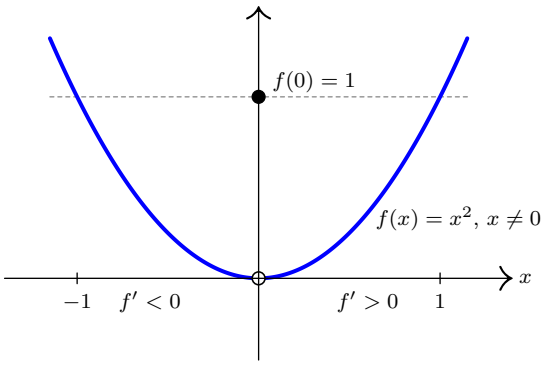


Figure 3: $f(x) = x^2$ for $x \neq 0$, $f(0) = 1$. The signs of f' match the minimum pattern, but f is not continuous at 0, and 0 is a strict local maximum.

The sign pattern of f' on the punctured neighborhood constrains only the relative order of f on the two sides; continuity at c is what ties that behavior to $f(c)$.

Remark 6.3 (Endpoints). The theorem is stated for interior points and gives no information at endpoints, where f' is defined (if at all) on only one side. For $f(x) = x$ on $[0, 1]$, f has a minimum at 0 and a maximum at 1, with no sign change of $f' \equiv 1$ available at either point. Endpoints must be examined by direct comparison of values, not by the sign of f' on two sides.

7 Concavity

Definition 7.1 (Convex and concave functions)**.** Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. We say f is *convex* (or *concave up*) on I if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) \qquad \forall x, y \in I,\ \forall t \in [0, 1].$$

We say f is *concave* (or *concave down*) on I if the reverse inequality holds. The inequalities are non-strict, so a linear function is both convex and concave.

Proposition 7.2 (Monotone derivative criterion)**.** *Let f be differentiable on an interval I. Then f is convex on I if and only if f' is nondecreasing on I.*

Proof. ($\Leftarrow$) Suppose f' is nondecreasing. Fix $x < y$ in I and $t \in (0, 1)$, and let $z = tx + (1-t)y$, so $x < z < y$. By the Mean Value Theorem there exist $\xi_1 \in (x, z)$ and $\xi_2 \in (z, y)$ with

$$\frac{f(z) - f(x)}{z - x} = f'(\xi_1), \qquad \frac{f(y) - f(z)}{y - z} = f'(\xi_2).$$

Since $\xi_1 < z < \xi_2$ and f' is nondecreasing, $f'(\xi_1) \leq f'(\xi_2)$, i.e.

$$\frac{f(z) - f(x)}{z - x} \leq \frac{f(y) - f(z)}{y - z}.$$

Using $z - x = (1-t)(y - x)$ and $y - z = t(y - x)$ and clearing denominators (both positive) yields, after rearranging,

$$f(z) \leq tf(x) + (1-t)f(y),$$

which is the convexity inequality at $z = tx + (1-t)y$.

($\Rightarrow$) Suppose f is convex and let $x < y$ in I. A standard consequence of the convexity inequality is that the difference quotient $\big(f(w) - f(x)\big)/(w - x)$ is nondecreasing in $w > x$; letting $w \to x^+$ and $w \to y^-$ shows $f'(x) \leq f'(y)$. □

Proposition 7.3 (Second-derivative criterion)**.** *If f is twice differentiable on an interval I and $f'' \geq 0$ on I, then f is convex on I; if $f'' \leq 0$ on I, then f is concave on I.*

Proof. If $f'' \geq 0$ on I, then for $x < y$ in I the Mean Value Theorem applied to f' gives $\xi \in (x, y)$ with $f'(y) - f'(x) = f''(\xi)(y - x) \geq 0$, so f' is nondecreasing on I. Convexity follows from the previous proposition. The concave case is analogous. □

Proposition 7.4 (Tangent-line characterization)**.** *If f is differentiable and convex on I, then for all $a, x \in I$,*

$$f(x) \geq f(a) + f'(a)(x - a).$$

If f is differentiable and concave on I, the reverse inequality holds.

Proof. Suppose f is convex, so f' is nondecreasing by the monotone derivative criterion. For $x > a$, the Mean Value Theorem gives $\xi \in (a, x)$ with $(f(x) - f(a))/(x - a) = f'(\xi) \geq f'(a)$, so $f(x) - f(a) \geq f'(a)(x - a)$. For $x < a$, the Mean Value Theorem on $[x, a]$ gives $\xi \in (x, a)$ with $(f(a) - f(x))/(a - x) = f'(\xi) \leq f'(a)$; multiplying by $a - x > 0$ and rearranging again gives $f(x) \geq f(a) + f'(a)(x - a)$. The case $x = a$ is trivial. The concave case follows by applying this to $-f$. □

The geometric content is that the graph of a convex function lies on or above every tangent line, and the graph of a concave function lies on or below every tangent line.

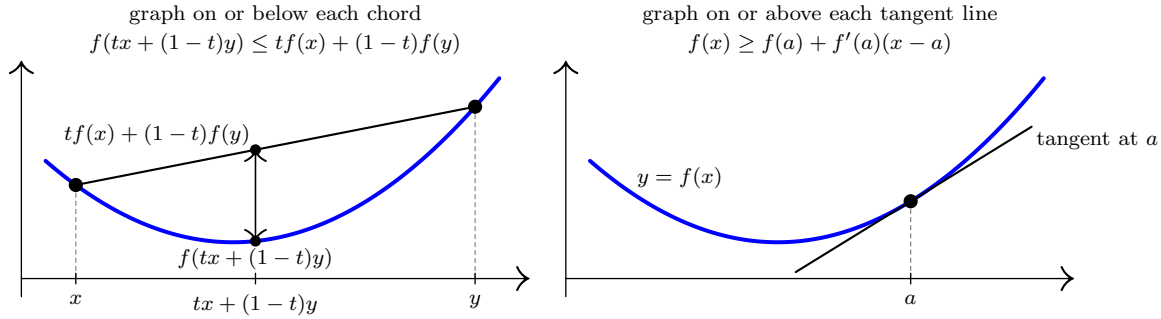


Figure 4: Two equivalent pictures of a concave up (convex) function: the graph lies on or below its chords, and, when f is differentiable, on or above its tangent lines.

Example 7.5. $f(x) = x^2$ has $f''(x) = 2 > 0$, so f is convex on $\mathbb{R}$; its graph lies above every tangent line. $f(x) = -x^2$ has $f''(x) = -2 < 0$, so f is concave on $\mathbb{R}$.

Definition 7.6 (Inflection point)**.** A point c is an *inflection point* of f if f is convex on one side of c and concave on the other.

Example 7.7. For $f(x) = x^3$, $f''(x) = 6x$, which is negative for $x < 0$ and positive for $x > 0$. So f is concave on $(-\infty, 0)$ and convex on $(0, \infty)$, and 0 is an inflection point. Note that 0 is not a local extremum of x^3: a change of concavity is a distinct phenomenon from an extremum.

8 The Second Derivative Test

Theorem 8.1 (Second derivative test)**.** *Let c be an interior point of the domain of f, suppose f' is defined on a neighborhood of c with $f'(c) = 0$, and suppose $f''(c)$ exists.*

1. *If $f''(c) > 0$, then f has a strict local minimum at c.*

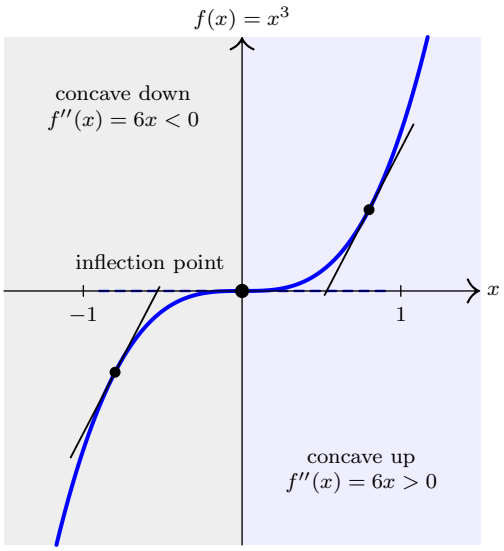


Figure 5: x^3 is concave down on $(-\infty, 0)$, where tangent lines lie above the graph, and concave up on $(0, \infty)$, where they lie below. At the inflection point 0 the tangent line crosses the graph.

2. *If $f''(c) < 0$, then f has a strict local maximum at c.*
3. *If $f''(c) = 0$, the test is inconclusive.*

Proof. Suppose $f''(c) > 0$. By definition, and using $f'(c) = 0$,

$$f''(c) = \lim_{x \to c} \frac{f'(x) - f'(c)}{x - c} = \lim_{x \to c} \frac{f'(x)}{x - c}.$$

Since this limit is positive, there exists $\delta > 0$ such that $f'(x)/(x - c) > 0$ for all x with $0 < |x - c| < \delta$. For $x \in (c - \delta, c)$, $x - c < 0$, forcing $f'(x) < 0$; for $x \in (c, c + \delta)$, $x - c > 0$, forcing $f'(x) > 0$. This is exactly the sign pattern $f' < 0$ on $(c - \delta, c)$ and $f' > 0$ on $(c, c + \delta)$, so the first derivative test gives a strict local minimum at c. The case $f''(c) < 0$ is symmetric.

For (3), x^4, $-x^4$, and x^3 (§11) all satisfy $f'(0) = 0 = f''(0)$ while exhibiting a strict local minimum, a strict local maximum, and no extremum respectively; hence no conclusion of the stated form can hold when $f''(c) = 0$. □

The proof shows the second derivative test is not an independent phenomenon: a nonzero $f''(c)$, together with $f'(c) = 0$, forces exactly the sign change of f' required by the first derivative test.

Remark 8.2. The stationary-point hypothesis $f'(c) = 0$ cannot be dropped. For $f(x) = x^2 + x$, $f''(0) = 2 > 0$, but $f'(0) = 1 \neq 0$ and 0 is not a local extremum (the minimum of this function is at $x = -1/2$). So

$$f''(c) > 0 \quad \not\Longrightarrow \quad c \text{ is a local minimum}$$

without the hypothesis $f'(c) = 0$.

9 Taylor's Theorem as a Local Model

Definition 9.1 (Little-o)**.** For a function g defined near 0 and $n \in \mathbb{N}$, we write $g(h) = o(h^n)$ as $h \to 0$ if

$$\lim_{h \to 0} \frac{g(h)}{h^n} = 0.$$

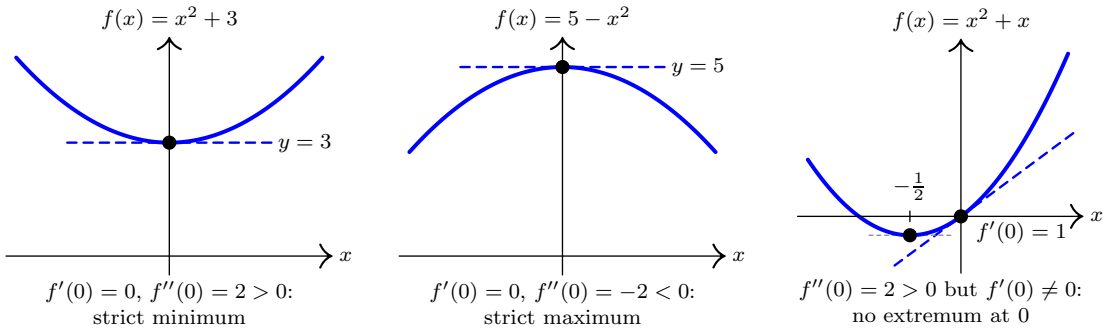


Figure 6: The second derivative test. At a horizontal tangent, $f'' > 0$ puts the graph above it and $f'' < 0$ below it; without $f'(c) = 0$, the sign of $f''(c)$ says nothing about an extremum at c.

Theorem 9.2 (Taylor, Peano form)**.** *Suppose $f^{(n-1)}$ exists on a neighborhood of c and $f^{(n)}(c)$ exists. Then*

$$f(c+h) = f(c) + f'(c)h + \frac{f''(c)}{2!}h^2 + \cdots + \frac{f^{(n)}(c)}{n!}h^n + o(h^n) \qquad \text{as } h \to 0.$$

We take this theorem as known; it is a standard result of one-variable calculus, proved by repeated application of L'Hôpital's rule or Cauchy's mean value theorem to the remainder. We use it to re-derive the two derivative tests as corollaries, with the sign arguments made fully rigorous via the definition of $o(h^n)$.

Corollary 9.3 (Fermat's theorem, again)**.** *If $f'(c) = 0$ fails, i.e. $f'(c) \neq 0$, then c is not an interior local extremum.*

Proof. By Taylor's theorem with $n = 1$, $f(c+h) - f(c) = f'(c)h + o(h)$. Let $\varepsilon = |f'(c)|/2 > 0$; there is $\delta > 0$ such that $|o(h)| < \varepsilon|h|$ for $0 < |h| < \delta$. Then for such h, $f(c+h) - f(c)$ has the same sign as $f'(c)h$, namely the sign of $f'(c)$ when $h > 0$ and the opposite sign when $h < 0$. Since these signs differ, $f(c+h) - f(c)$ takes both signs on every punctured neighborhood of c, so c is not a local extremum. □

Corollary 9.4 (Second derivative test, again)**.** *Suppose $f'(c) = 0$ and $f''(c)$ exists. If $f''(c) > 0$, c is a strict local minimum; if $f''(c) < 0$, c is a strict local maximum.*

Proof. By Taylor's theorem with $n = 2$ and $f'(c) = 0$,

$$f(c+h) - f(c) = \frac{f''(c)}{2}h^2 + o(h^2).$$

Suppose $f''(c) > 0$ and let $\varepsilon = f''(c)/4 > 0$. Choose $\delta > 0$ such that $|o(h^2)| < \varepsilon h^2$ for $0 < |h| < \delta$. Then for such h,

$$f(c+h) - f(c) \geq \frac{f''(c)}{2}h^2 - \varepsilon h^2 = \frac{f''(c)}{4}h^2 > 0.$$

So $f(c+h) > f(c)$ throughout the punctured neighborhood $0 < |h| < \delta$, a strict local minimum. The case $f''(c) < 0$ is symmetric. □

Unlike the earlier proof from the limit definition of $f''(c)$, this argument makes the role of the remainder explicit: the conclusion depends on the $o(h^2)$ term being eventually dominated by the quadratic term, which is exactly the content of the little-o definition.

10 Examples of the Second Derivative Test

Example 10.1. For $f(x) = x^2 + 3$, $f'(x) = 2x$ and $f''(x) = 2$. At $c = 0$, $f'(0) = 0$ and $f''(0) = 2 > 0$, so 0 is a strict local minimum.

Example 10.2. For $f(x) = 5 - x^2$, $f'(x) = -2x$ and $f''(x) = -2$. At $c = 0$, $f'(0) = 0$ and $f''(0) = -2 < 0$, so 0 is a strict local maximum.

11 The Case $f''(c) = 0$

When $f'(c) = 0 = f''(c)$, all three outcomes are possible.

Example 11.1. $f(x) = x^4$ has $f'(0) = f''(0) = 0$, and $x^4 \geq 0 = f(0)$ for all x, so 0 is a strict local minimum.

Example 11.2. $f(x) = -x^4$ has $f'(0) = f''(0) = 0$ and 0 is a strict local maximum, by the same argument applied to $-x^4$.

Example 11.3. $f(x) = x^3$ has $f'(0) = f''(0) = 0$, yet 0 is neither a maximum nor a minimum (§2); it is instead an inflection point, since $f''(x) = 6x$ changes sign at 0 (§7).

Since all three outcomes occur under identical data $f'(c) = f''(c) = 0$, no theorem of the form "$f'(c) = 0 = f''(c)$ implies ..." concerning the extremum classification can be true; this is precisely the content of "inconclusive."

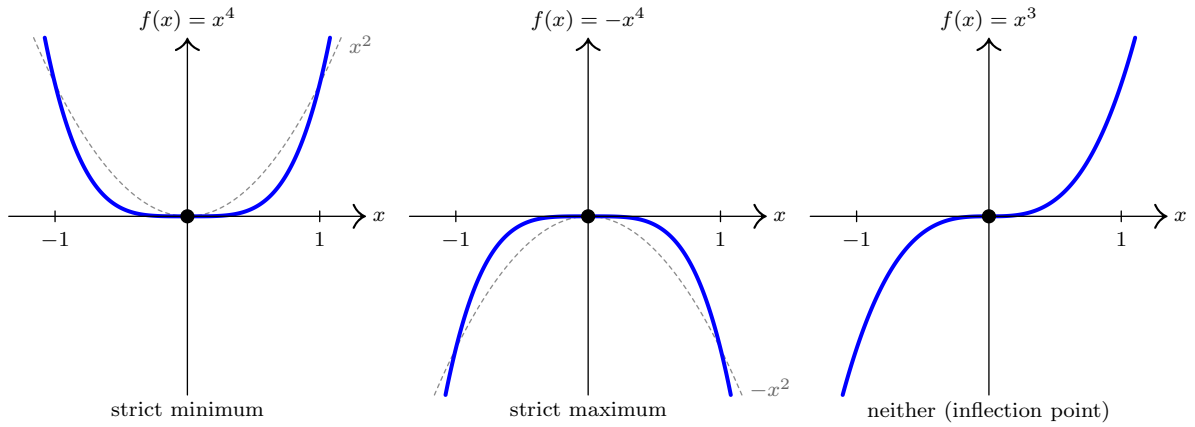


Figure 7: Three functions with $f'(0) = f''(0) = 0$ and three different outcomes, which is why the second derivative test is inconclusive when $f''(c) = 0$.

12 An Extremum with No Useful Higher-Order Data

Example 12.1. $f(x) = |x|$ has a strict local minimum at 0, but neither $f'(0)$ nor $f''(0)$ exists: the second derivative test is a sufficient condition, never a necessary one.

Example 12.2. $f(x) = |x|^{3/2}$ is differentiable at 0 with $f'(0) = 0$ (indeed $f'(x) = \frac{3}{2}\operatorname{sgn}(x)\sqrt{|x|} \to 0$ as $x \to 0$), and 0 is a strict local minimum since $|x|^{3/2} \geq 0 = f(0)$. But $f''(x) = \frac{3}{4}|x|^{-1/2} \to \infty$ as $x \to 0$, so $f''(0)$ does not exist and the second derivative test cannot be applied.

13 Necessary versus Sufficient Conditions

For an interior point c at which f is differentiable, Fermat's theorem gives the necessary condition

$$c \text{ is a local extremum } \implies f'(c) = 0,$$

which is not sufficient (x^3). Under the stated neighborhood hypotheses, a sign change of f' across c is sufficient for an extremum (first derivative test), and $f'(c) = 0$ together with $f''(c) \neq 0$ is sufficient (second derivative test), but neither test is necessary: an extremum can occur where f' fails to exist ($|x|$) or where $f''(c)$ fails to exist ($|x|^{3/2}$) or vanishes (x^4). The logical structure is asymmetric: differentiable extrema are always critical points, but not every classification tool applies to every critical point.

14 Strict versus Non-Strict Extrema

The derivative tests as stated detect *strict* extrema, via an actual sign change of f'. Non-strict extrema need not exhibit such a change.

Example 14.1. The constant function $f \equiv 0$ has $f(x) = f(c)$ for all x, c, so every point is simultaneously a (non-strict) local maximum and minimum, yet $f' \equiv 0$ exhibits no sign change anywhere.

Example 14.2. Let

$$f(x) = \begin{cases} x^2, & x < 0, \\ 0, & 0 \leq x \leq 1, \\ (x-1)^2, & x > 1. \end{cases}$$

Every point of $[0, 1]$ is a (non-strict) local minimum, but an interior point of $(0, 1)$ has $f' \equiv 0$ on a full neighborhood, not a sign change from negative to positive.

The first derivative test, as stated, is a sufficient condition for a *strict* extremum; it should not be read as characterizing every non-strict extremum.

15 Endpoints and Constrained Domains

For $f : [a, b] \to \mathbb{R}$, an absolute extremum can occur at an interior critical point, at an interior point of non-differentiability, or at either endpoint a or b — points to which Fermat's theorem and the two derivative tests, stated for interior points, do not apply. The standard method for locating an absolute extremum on $[a, b]$ is therefore to evaluate f at every interior critical point and at both endpoints, and compare the resulting values directly, rather than to rely on derivative sign patterns at the endpoints.

Example 15.1. $f(x) = x$ on $[0, 1]$ has no interior critical point, yet $\min_{[0,1]} f = f(0) = 0$ and $\max_{[0,1]} f = f(1) = 1$, found by direct comparison of the two endpoint values.

16 Higher-Order Derivative Tests

Theorem 16.1 (nth derivative test)**.** *Suppose* $f'(c) = f''(c) = \cdots = f^{(n-1)}(c) = 0$ *and* $f^{(n)}(c) \neq 0$*, where* $f^{(n-1)}$ *exists on a neighborhood of* c*.*

1. *If n is even and $f^{(n)}(c) > 0$, c is a strict local minimum.*
2. *If n is even and $f^{(n)}(c) < 0$, c is a strict local maximum.*
3. *If n is odd, c is not a local extremum.*

Proof. By Taylor's theorem (Peano form), since all derivatives up to order $n-1$ vanish at c,

$$f(c+h) - f(c) = \frac{f^{(n)}(c)}{n!}h^n + o(h^n).$$

Let $\varepsilon = |f^{(n)}(c)|/(2 \cdot n!)$; choose $\delta > 0$ such that $|o(h^n)| < \varepsilon |h|^n$ for $0 < |h| < \delta$. For such h, $f(c+h) - f(c)$ has the same sign as $f^{(n)}(c)h^n/n!$.

If n is even, $h^n > 0$ for every $h \neq 0$, so this sign equals the sign of $f^{(n)}(c)$ on both sides of c, giving (1) and (2) exactly as in the second-derivative case.

If n is odd, h^n has the sign of h, so $f(c+h) - f(c)$ has opposite signs for $h > 0$ and $h < 0$; hence c is not a local extremum. □

Example 16.2. For $f(x) = x^4$, the first nonvanishing derivative at 0 is $f^{(4)}(0) = 24 > 0$ with $n = 4$ even, giving a local minimum, matching §11.

Example 16.3. For $f(x) = x^3$, the first nonvanishing derivative at 0 is $f'''(0) = 6$ with $n = 3$ odd, so 0 is not a local extremum, matching §2.

17 Smooth Is Not Analytic: A Counterexample

The nth derivative test presupposes some n with $f^{(n)}(c) \neq 0$. This can fail even for an infinitely differentiable function.

Example 17.1. Let

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

A standard computation shows f is infinitely differentiable at 0 with $f^{(n)}(0) = 0$ for every $n \geq 1$. Yet $f(x) > 0 = f(0)$ for every $x \neq 0$, so 0 is a strict local minimum. No finite-order derivative test can detect this, since every derivative at 0 vanishes: the Taylor series of f at 0 is identically 0, while f itself is not identically 0 on any neighborhood of 0. This is the standard example separating smooth (C^∞) functions from analytic ones.

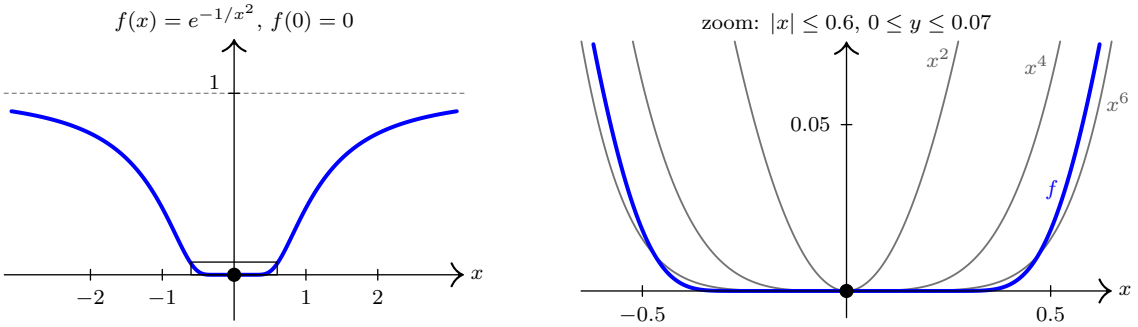


Figure 8: $f(x) = e^{-1/x^2}$, $f(0) = 0$. Every derivative of f at 0 vanishes, so f is flatter at 0 than any power of x (right: it lies below even x^6 near 0), yet 0 is a strict minimum.

18 Comparison of the Two Tests

The first derivative test uses information about f' on a punctured neighborhood of c and requires only continuity, not differentiability, at c itself. The second derivative test uses only the two numbers $f'(c)$ and $f''(c)$; it presupposes more regularity (f' on a neighborhood of c, and $f''(c)$), but is often faster to check. Every point the second derivative test classifies is also classified by the first, since its proof produces the required sign change of f'; the converse fails. For $f(x) = x^4$, the second derivative test is inconclusive ($f''(0) = 0$), while $f'(x) = 4x^3$ changes sign from negative to positive at 0, so the first derivative test gives a strict local minimum. For $f(x) = |x|$, the second derivative test cannot even be stated, since $f'(0)$ does not exist, while the first derivative test applies directly.

19 A Logical Checklist for Extremum Problems

1. Determine the domain, including endpoints and excluded points, before differentiating.
2. Locate the candidate points: solutions of $f'(x) = 0$, points where f' fails to exist, and (for absolute extrema) the endpoints.
3. At each interior candidate, check whether the first derivative test's hypotheses hold, i.e. whether f' has a fixed sign on each side of c.
4. If $f'(c) = 0$, compute $f''(c)$; a nonzero value classifies c immediately by the second derivative test.
5. If $f''(c) = 0$, draw no conclusion from this alone; return to the sign of f' directly, or to a higher-order derivative test if its hypotheses hold.
6. For an absolute extremum on a closed interval, compare the values of f at all candidates and both endpoints directly.

20 Common Logical Errors

1. *"$f'(c) = 0$ implies c is an extremum."* False; $f(x) = x^3$ at $c = 0$.
2. *"$f''(c) > 0$ implies c is a minimum."* False without $f'(c) = 0$; $f(x) = x^2 + x$ at $c = 0$.
3. *"$f''(c) = 0$ implies c is an inflection point."* False; $f(x) = x^4$ at $c = 0$ has $f''(0) = 0$ but no change of concavity there, since $f''(x) = 12x^2 \geq 0$ on both sides.
4. *"If the second derivative test fails, there is no extremum."* False; $f(x) = x^4$ at $c = 0$.
5. *"Every extremum occurs where $f'(c) = 0$."* False; $f(x) = |x|$ at $c = 0$, and any endpoint extremum.
6. *"A critical point is an extremum."* False in general; a critical point is only a candidate.

21 Summary

Hypothesis	Conclusion	Source
$f'(c) = 0$	candidate only	Fermat (necessary, not sufficient)
$f' : - \to +$ across c	strict local minimum	first derivative test
$f' : + \to -$ across c	strict local maximum	first derivative test
f' fixed sign across c	no local extremum	first derivative test
$f'' \geq 0$ on I	f convex on I	concavity criterion
$f'' \leq 0$ on I	f concave on I	concavity criterion
f'' changes sign across c	inflection point	concavity change
$f'(c) = 0,\ f''(c) > 0$	strict local minimum	second derivative test
$f'(c) = 0,\ f''(c) < 0$	strict local maximum	second derivative test
$f'(c) = 0,\ f''(c) = 0$	inconclusive	$- - -$

22 Conclusion

The derivative tests are theorems, not definitions: they relate local sign and vanishing behavior of derivatives to the local order structure of f, under hypotheses that must be verified rather than assumed. The first derivative test is fundamentally a statement about monotonicity, proved from the Mean Value Theorem; the second derivative test is a sufficient condition, at a stationary point, that forces the sign change required by the first, provable either from the limit definition of $f''(c)$ or from Taylor's theorem with Peano remainder. Concavity is a logically separate notion — monotonicity of f' rather than sign of f or f' — and an inflection point is a change of concavity, not an extremum.

Every implication proved here runs in one direction only. $f'(c) = 0,\ f''(c) > 0$ is sufficient, not necessary, for a strict local minimum: a minimum may occur with $f''(c) = 0$ (x^4), with $f''(c)$ undefined ($|x|^{3/2}$), with $f'(c)$ undefined ($|x|$), or with every derivative at c vanishing while f is still smooth (e^{-1/x^2}). Correct use of these tests therefore requires checking, at each step, exactly which hypothesis is being invoked and whether it actually holds — not merely computing derivatives and reading off a sign.